Class Meeting Handout 12

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Hammett σ substituent parameters work well in many systems, but in others they refuse to work well. One thing that these substituents have in common is the ability to accept or donate electrons via resonance. Today, we will explore substituent parameters that include these extra resonance effects.

This document was produced using the LaTeX typesetting language with the Tufte-handout document class. Chemical diagrams were created in *ChemDoodle*. Plots were typeset using *MatPlotLib* and *Python* tools. The plot was then further edited in *Affinity Designer*.

LFER Part 2: Substituents with More Punch

As you read the textbook[1] and try the suggested problems, here are some things that we will be discussing in class.

[1] Read of ch. 8.3 again.

Discussion: Brown-Okamoto Parameters – 20 minutes

Were we not quite satisfied? In the previous class meeting, we observed Hammett plots with obvious outliers. Was there a reason?

1. Re-evaluate the Hammett plots for hydrolysis of benzyl chloride, diphenylmethylene chloride and trityl acetates from the previous class meeting[2] using the Brown-Okamoto substituent parameter, σ^+. While we are doing this we will consider the following questions. Be prepared to answer them in class.

   (a) What are the σ^+ substituent parameters? How were they measured? What information do they provide?

   (b) When would you use σ^+ substituent parameters rather than the Hammett σ substituent parameters?

   (c) Which plots from the previous meeting were improved when using σ^+? Does this fit with your understanding of σ^+ values?

   (d) Many substituents do not have a unique σ^+ value. What should we use for substituent parameters in these cases?

   (e) No meta-substituents have unique values for σ^+. Why?

[2] See the class meeting handout from the previous meeting for the data tables. The moodle site has a supplemental handout with these plots available.

You can revisit the plots and change the substituent parameters by returning to the *Python* notebooks on Google Colab.

Table 4 plot from previous handout: https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/Class_14/Table_4_Plots.ipynb

Table 5 plot from previous handout: https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/Class_14/Table_5_Plots.ipynb

Table 6 plot from previous handout: https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/Class_14/Table_6_Plots.ipynb

Table 7 plot from previous handout: https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/Class_14/Table_7_Plots.ipynb

Exercise: Applying Brown-Okamoto Parameters – 20 minutes

2. Consider the discussion presented in the following pages[3] for the nucleophilic attack of phenoxide ions on epoxides. Be prepared to answer the following questions.

   (a) The kinetic data seems to correlate better with σ^- values. What are σ^- values? How were they measured? What information do they provide?

   (b) When would you use σ^- substituent parameters rather than σ or σ^+ parameters?

[3] See the "Exploration #1: Substituent Constants for Resonance Effects" section below.

Exercise: Applying Brown-Okamoto Parameters – 10 minutes

3. Describe when it is appropriate to use σ, σ^+ or σ^- as the basis for correlating structure with reactivity. Could there be reactions that have only partial sensitivity to resonance effects? Are there parameters that fit in-between σ and σ^+/σ^-? Consider the discussion at the end of this document about the Yukawa-Tsuno approach[4] and apply it to the hydrolysis of trityl acetates discussed in the previous meeting.

[4] See the "Exploration #2: Partial Resonance Contributions" section below.

You can examine how to analyze the data in table 7 of the previous handout using the Yukawa-Tsuno equation and optimize the r value using the *Python* notebook at https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/Class_14/Table_7_Plots_with_YukawaTsuno.ipynb

Learning Goals

Through the suggested textbook reading, problems and participating in this class discussion, students will…

- have gained further experience in resonance effects and electron delocalization.
- be able to define the parameters in the Brown-Okamoto (σ^+ & σ^-) and the Yukawa-Tsuno LFER systems.
- be able to predict when σ, σ^+, or σ^- substituent parameters should be used.
- **Computer Skill:** have again used an interactive *Python* notebook to interpret and analyze experimental data from the literature.

A Look Ahead

We have shown that structural differences can require different substituent constants and that the connection for resonance electron sharing is not "all or nothing." In the next class meeting, we will explore what happens when the electronic effect of substituents results in an entirely new mechanism. We will cross that line together.

A Look Ahead to Exploration

You will see two examples of very short explorations of topics in the following few pages. Observe how these explorations are based on analyzing the reported literature data. Soon you will be asked to explore a publication with a critical spirit (you will be asked to be nasty.) The two explorations that follow are good examples (albeit small) of what you will be asked to produce at the end of this course.

Exploration #1: Substituent Constants for Resonance Effects

The Hammett equation relates the "score" for an electron-withdrawing or electron-donating group to the log of the observed rate or rate constant. The Hammett scores are based on the observed change in the pK_a for substituted benzoic acids. These are designated as σ_m and σ_p for each substituent. Some reactions have electronic effects that are far greater than described in the Hammett σ substituent parameters.

If the reaction centre can connect via resonance with the substituent, there will be a far greater influence of the group. Substituents that are *meta* to the reaction centre cannot connect via resonance, but *para* substituents can. If a group can accept or donate electrons by resonance, we will need to replace Hammett's σ_p values with the σ_p^- and σ_p^+ values of Brown and Okamoto.[5] I found some data for nucleophilic ring-opening of epoxides by phenoxides in an ancient piece of work[6] that predates the field of physical organic chemistry[7] itself.

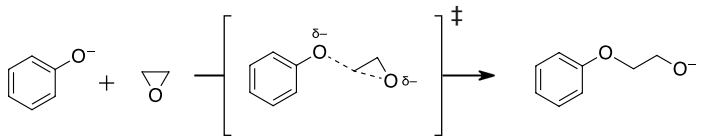


Figure 1: Ring-opening of epoxide by phenoxides.
←

The oxygen nucleophile of phenoxide would be able to connect via resonance with electron acceptors in the *para* position. Substituents that can accept electrons by resonance should have a much greater electron-withdrawing effect compared to the case of benzoic acid. When the electronic structure resembles a phenol, we must use substituent constants designed for the situation.

We will plot the $\log k_{obs}$ *vs.* σ^- values from the data tables. The open circles are points using the Hammett σ_p values, and the closed circles are points using σ_m and σ_p^- values. When including Brown-Okamoto parameters in Hammett plots, we generally refer to the axis as σ^+ or σ^-. We use σ_p and σ_m values for substituents where resonance has no significant effect, and use the σ_p^+ or σ_p^- values when there is a resonance effect.

Table 1: Hammett and Brown-Okamoto substituent constants.↓

Substituent	σ	σ^+	σ^-
p-OCH_3	−0.27	−0.78	−0.26
p-CH_3	−0.17	−0.31	−0.17
m-CH_3	−0.07		
p-C_6H_5	−0.01	0.02	−0.18
H	0.00		
p-Cl	0.23	0.11	0.19
m-Cl	0.37		
m-NO_2	0.71		
p-NO_2	0.78		1.27
p-CN	0.66		1.00
m-CN	0.56		
p-$N_2C_6H_5$	0.31	−0.19	0.45

Table 2: Rate constants for nucleophilic attack of substituted phenols on epoxide.[6]↓

Substituent	$k_{obs}\ hr^{-1}$
p-NO_2	0.013
p-CN	0.026
m-NO_2	0.032
m-CN	0.044
p-$N_2C_6H_5$	0.050
m-Cl	0.101
p-Cl	0.108
H	0.205
m-CH_3	0.256
p-CH_3	0.257

Exploration #2: Partial Resonance Contributions

The Yukawa-Tsuno equation[8] combines the "scores" for both the mostly inductive substituent constants of Hammett with the resonance substituent constants of Brown & Okamoto. While most Hammett plots fit well when using one set of substituent parameters or the other, there are cases where the best fit is when we use a weighted average of the two sets.

Yukawa and Tsuno proposed a plot using a weighting factor, r, that is adjusted to achieve the best line fit. Plots where r is significantly different

[5] "A Quantitative Treatment for Electrophilic Reactions of Aromatic Derivatives", Y. Okamoto and H.C. Brown, *J. Org. Chem.*, **1957**, *22*, 485-494. https://doi.org/10.1021/jo01356a003 and "Substituent Constants for Aromatic Substitution", Herbert C. Brown and Y. Okamoto, *J. Am. Chem. Soc.* **1957**, *79*, 1913-1917. https://doi.org/10.1021/ja01565a039

[6] "The Velocities of Combination of Sodium Derivatives of Phenols with Olefine Oxides.", David Runciman Boyd and Ernest Robert Marle., *J. Chem. Soc., Trans.*, **1914**, *105*, 2117-2139. https://doi.org/10.1039/CT9140502117

[7] Louis P. Hammett (1894-1987) did not publish his famous textbook until 1940.

[8] "Sigma Constants in Electrophilic Reactions and their Intercorrelation." Y. Yukawa, Y. Tsuno, *Bull. Chem. Soc. Japan*, **1959**, *32*, 965-971. https://doi.org/10.1246/bcsj.32.965

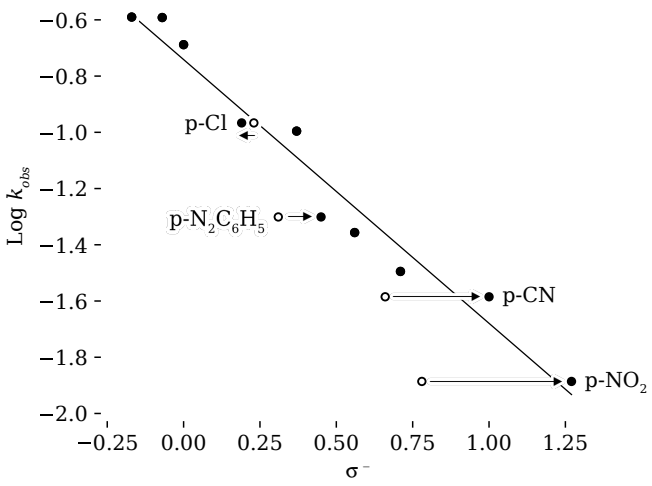


Figure 2: $\log k_{obs}$ *vs.* σ values for ring-opening of epoxide by phenoxides. ←
Open circles are Hammett σ values, closed circles are σ^- values. The slope of the line is -0.90.

This plot is generated from data set in table 2 and 1 and can be investigated by using the *Python* notebook available via Colab at https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/Class_15/15-BrownOkamoto_Python_Notebook.ipynb.

than 1 or 0 indicate a transition state that has partial resonance participation by the substituent.

The acid hydrolysis of phenylacetals proceeds through a S_N1 mechanism.[9] The protonated leaving group is assisted in departure by the donation of electrons from the adjacent oxygen into the forming p-orbital. Oxygen groups are highly stabilizing for adjacent carbocations, and this direct electron donation greatly reduces the importance of the resonance contribution of the Hammett substituents.

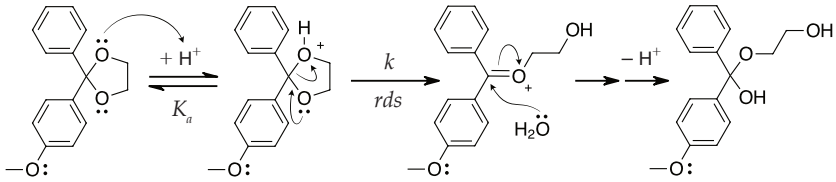


Figure 3: Acid hydrolysis of an acetal. ←

We would expect the σ^+ parameter series to fit best when a carbocation is attached to the phenyl ring. However, in this case it seems that neither the σ^+ series nor the Hammett σ series fits.[10] This is because the *p*-OCH_3 substituent has a large difference between its σ^+ and σ values and neither option lands on the line being telegraphed by the other data points. If only there was a set of σ values for partial resonance donors. That is exactly what Yukawa and Tsuno did. They created a sliding scale between σ and σ^+ using their equation.

The r value represents this scale. When $r = 1$ we have the σ^+ series and when $r = 0$ we have the σ series. In this reaction, the data fits best when we use a series of values in between σ^+ and σ. But how do we find the optimal value for r that gives the best line fit?

Using a literature example,[9] we will explore the Yukawa-Tsuno plot for phenylacetal hydrolysis. The rate data for the hydrolysis reaction is presented

The Yukawa-Tsuno Equation.↓

$$\log \frac{k_X}{k_H} = \rho \cdot [\sigma + r(\sigma^+ - \sigma)]$$

[9] "Substituent Effects in the Acid-Catalyzed Hydrolysis of Cyclic Acetals of Benzophenones", U. Shinya, O. Takumi, N. Toshikazu, *Bull. Chem. Soc. Japan*, **1986**, *59*, 305-306. https://www.journal.csj.jp/doi/10.1246/bcsj.59.305

The reaction scheme.↓

$$A + H^+ \underset{K_a}{\rightleftharpoons} AH^+ \xrightarrow{k} P$$

The rate law.↓

$$\nu = k_{obs}[A], \text{ and } k_{obs} = \frac{[H^+]}{K_a + [H^+]}k$$

[10] The r^2 for the fit with Hammett σ substituent constants is actually good. It is 0.95. Using the σ^+ series of Brown and Okamoto give $r^2 = 0.92$, also an adequate (but not great) fit. Using the Yukawa-Tsuno approach, we can get $r^2 = 0.97$ (see figure 5.) This is a better fit. All of this is controlled by one single data point, the p-OCH_3 substituent. Does that seem a bit sketchy to you?

in table 3 and substituent constants are presented in table 1.

A Yukawa-Tsuno plot was constructed using the Hammett $\sigma_{m,p}$ constants and the Brown-Okamoto σ^+ constants. See figure 4.

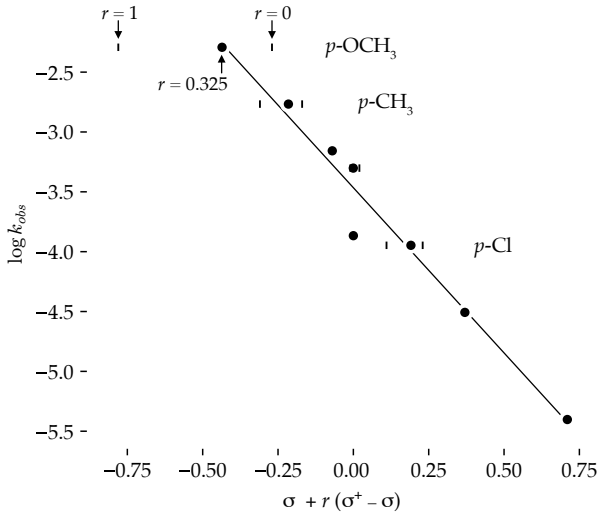


Figure 4: $\log k_obs$ *vs.* σ values for acid hydrolysis of phenylacetals.
←
Vertical slashes represent the edge cases when $r = 0$ (Hammett) and $r = 1$ (Brown-Okamoto). The circular markers represent $r = 0.325$. the slope is $\rho = -2.8$.

This plot is generated from data set in tables 3 and 1 and can be investigated by using the *Python* notebook available via Colab at https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/Class_15/15-YukawaTsunoFitNotebook.ipynb.

Table 3: Rate constants for hydrolysis of substituted diphenylacetals.↓

Substituent	k_{obs} /$10^{-4} s^{-1}$	$\log k_{obs}$
p-OCH_3	51.0	−2.29
p-CH_3	17.1	−2.77
m-CH_3	6.96	−3.16
p-C_6H_5	4.99	−3.30
H	1.36	−3.87
p-Cl	1.13	−3.95
m-Cl	0.311	−4.51
m-NO_2	0.0396	−5.40

Examine the plot and observe how the two substituents that have the greatest difference between the values of σ and σ^+ fall well to the right or left of the best fit when $r = 0$ and $r = 1$. If we mix the σ and σ^+ values with $r = 0.325$, we observe the best fit as measured by the correlation coefficient r^2 (not to be confused with the Yukawa-Tsuno r factor).[11]

How do we find the best value for the r factor? One method would be to simply write a program to perform least-squares line fits across a series of values for r. The result for such an approach is shown in figure 5. I then used the `minimize_scalar` function in the `SciPy.stats` library in a *Jupyter* notebook[12] to determine the exact value of r that resulted in the largest value for the correlation coefficient, r^2. I used $-r^2$ rather than r^2 so that a maximum score would be a minimum in the plot.

Take note that the Yukawa-Tsuno r-value is determined mostly due to the influence of the one or two points on the plot that are most changed as we change r. The r factor is expressed with three significant figures here. Is that justified, given its dependance on a single data point? Also, if your plot does not include substituents with a large variance between the σ and σ^+ constants, then it will not effectively reveal partial resonance effects. Often the Yukawa-Tsuno *r*-value is set by the p-OCH_3 substituent alone.

[11] Throughout this document keep in mind that "r" refers to the Yukawa-Tsuno resonance factor and "r^2" refers to the Pearson correlation coefficient

[12] See the accompanying *Jupyter* notebook for the documentation of the code and methodology

Figure 5: Optimization of the r-value as scored by the Pearson correlation coefficient, $-r^2$.↓

